Idea Submission for AI Unlocked

Name of your Institute: IIT KHARAGPUR
Challenge Area: AI Study Buddy
Your Team Name: Alpha Gradient
Registered Email ID of your Team Lead: priyanshu.24@kgpian.iitkgp.ac.in

Project Title

StepWise AI – Multilingual Intelligent Study Buddy

The Team

We are a team of students from IIT Kharagpur who frequently rely on handwritten notes and step-by-step problem solving while studying for academic courses and competitive examinations. Through our own learning experiences, we observed that students often get stuck at intermediate steps of a solution, and existing AI tools usually provide final answers rather than guidance.

We decided to work together for AI Unlocked to build an AI-powered study buddy that understands handwritten reasoning in real time, validates steps, and provides hints instead of direct answers—closely mimicking how a human tutor assists students during problem solving.

The Concept

Students often understand the theory behind a concept but struggle while solving problems step-by-step, especially in subjects such as mathematics, physics, and engineering. Most existing AI tools either require typed input or generate complete solutions, which limits active learning. Additionally, handwritten notes are messy, difficult to revise, and lack real-time feedback.

Our project introduces a **digital scribble pad (writing pad)** where students can write using a stylus or pointing device. Each step written by the user is captured and analyzed by AI to determine whether it is logically correct or incorrect. If the student gets stuck or makes a mistake, the AI provides **contextual hints instead of final answers**.

The system also supports:

- **Copilot-style next-line suggestions** while writing
- **Autocorrect and clarity improvements** for handwritten notes
- **Multilingual explanations**, allowing students to receive hints and feedback in English or Indian languages such as Hindi

Over time, aggregated anonymized data can also help identify common error patterns across students, which can be valuable for educators.

We explicitly address the possibility of incorrect AI-generated feedback using the following methods:

1. **Low Temperature Responses**
   The language model is configured with a low temperature (e.g., 0.1) to ensure deterministic, logical, and factual responses rather than creative outputs.

2. **Step-by-Step Validation**
   The model is instructed to internally verify each step before responding and only comment on the current step rather than solving the entire problem.
3. **Grounding Using External Tools**
   For factual information or constants, the system can be grounded using external search tools to reduce hallucinations.

Target Audience or Market

Primary Audience

- K-12 students
- Undergraduate students
- Competitive exam aspirants (JEE, NEET, GATE)
- Students who rely on handwritten problem solving

Secondary Audience

- Coaching institutes
- Educational institutions seeking analytics on student learning patterns

India's EdTech market is rapidly growing, and with increasing adoption of tablets and digital pens, a handwritten AI tutoring system has significant market potential.

Personas

- **Rohan – The Student**
  A 17-year-old JEE aspirant who understands concepts but makes small algebraic and sign errors. StepWise AI detects mistakes instantly and provides hints, saving time and frustration.
- **Ms. Iyer – The Teacher**
  A mathematics instructor who uses aggregated analytics to identify that many students are making errors at the same step, allowing targeted teaching interventions.

How it works:

Writing Pad Workflow

1. The user writes on a digital whiteboard using a stylus or mouse.
2. An inactivity timer (e.g., 5 seconds) triggers a snapshot of the writing.

3. The snapshot is processed using handwriting OCR.

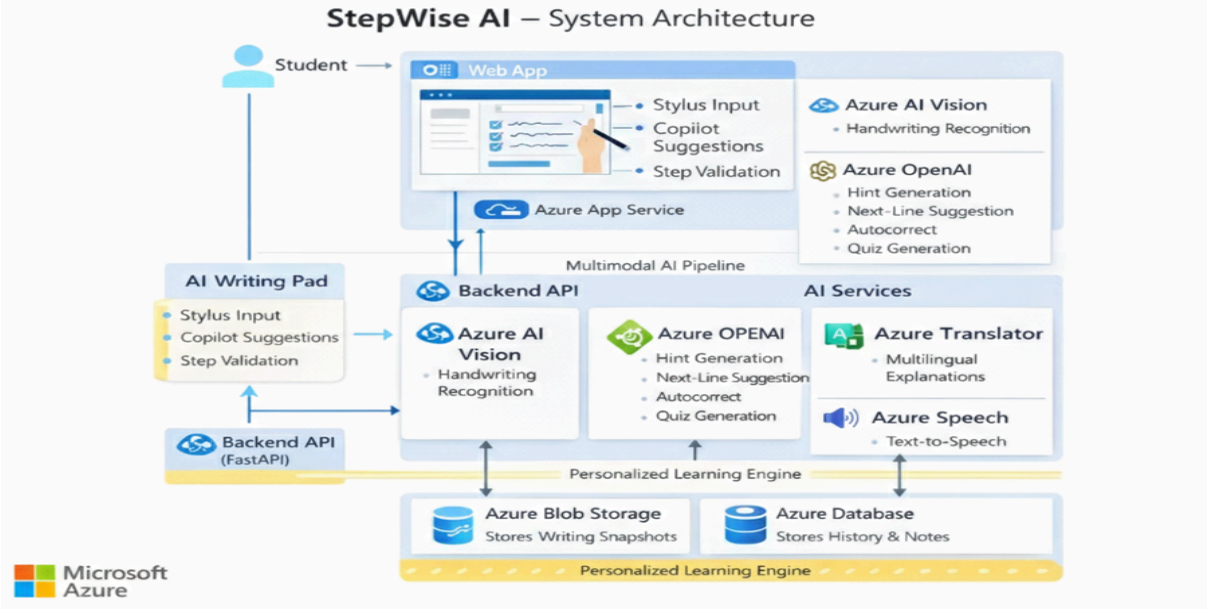
4. The extracted text is sent to a multimodal language model along with prior steps.

5. The AI:

    - validates the current step
    - suggests corrections if needed
    - provides hints if the student is stuck
    - suggests the next logical step (Copilot-style)
    - optionally translates feedback into the user's preferred language

Additional Features

- Upload handwritten notes as images
- Generate quizzes from notes
- Convert explanations to speech
- Multilingual explanations and hints



Core Technologies

- Azure OpenAI (GPT-4 class models)
- Azure AI Vision (handwriting OCR)
- Azure Translator (multilingual support)
- Azure Speech Services (text-to-speech)
- Azure App Service
- Azure Blob Storage
- Azure Cosmos DB / Table Storage
- Web-based writing canvas (React / similar)

The Business Plan

Freemium Model: Basic step validation and hints are free.
Premium Features:

- advanced explanations
- exam-mode practice
- detailed analytics

B2B Licensing:

- coaching institutes
- educational dashboards for error-pattern analysis